

Ankit Chaubey

Machine Learning Systems Researcher & Engineer

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EDUCATION

Indian Institute of Technology (IIT) Guwahati2025 – Present

Master of Technology (M.Tech) in Robotics & Artificial Intelligence

Research focus: From-scratch Transformer architectures, Rotary Position Embeddings (RoPE), KV-cache optimization, deep generative diffusion models, and sequential recommendation systems.

Dehradun Institute of Technology (DIT University)2020 – 2024

Bachelor of Technology (B.Tech) in Computer Science & Engineering

First Class with Distinction. Core coursework in Data Structures, Algorithms, Computer Systems, and Operating Systems.

FLAGSHIP ML SYSTEMS PROJECTS (BUILT & EVALUATED FROM SCRATCH)

LLM From Scratch in PyTorchPyTorch, Python, CUDA, C++May 2026 – Aug. 2026

A GPT-style transformer built from tokenization to RLHF, implementing RoPE, RMSNorm, SwiGLU, and KV-cache with first-principles CUDA optimization.

- **Rotary Position Embeddings (RoPE) over Learned Absolute Positional Embeddings:** Learned absolute embeddings fail to generalize beyond the training context window and lose relative token distances. RoPE injects relative positional information directly into the dot-product attention score ($q_m^T k_n$) by rotating query and key vectors by angles proportional to their positions m and n .
- **RMSNorm over LayerNorm:** LayerNorm calculates both the mean and variance of inputs across the hidden dimension: $LN(x) = (x - \mu) / \sigma * \gamma + \beta$. RMSNorm hypothesizes that the scaling property of LayerNorm provides 99% of the stabilization benefit, discarding the mean calculation: $RMSNorm(x) = x / \sqrt{\text{mean}(x^2) + \epsilon} * \gamma$.
- **Measured Metric:** Pre-allocated static KV-cache buffers eliminate dynamic GPU memory fragmentation and realloc pauses during autoregressive token decoding. (84.6 tok/s @ 2048 ctx vs 16.8 tok/s (naive tensor concat)).

RAGDoctor: RAG Failure Diagnosis & Auto-Remediation PlatformPython, PyTorch, FAISS, HuggingFaceJul. 2026 – Oct. 2026

Automated diagnostic engine for Retrieval-Augmented Generation that isolates retrieval, context-packing, and generation failures using multi-stage execution tracing and controlled perturbations.

- **Hybrid Sparse (BM25) + Dense (FAISS) with Reciprocal Rank Fusion (RRF):** Dense retrievers fail on exact entity codes, acronyms, and product serial numbers due to embedding compression. Sparse BM25 excels at exact keyword matching but fails on synonyms. Merging them via RRF avoids fragile score normalization heuristics.

- **Cross-Encoder Reranking over Candidate Pool Top-50:** Bi-encoders compute vector embeddings independently, precluding cross-token attention between query and passage. A Cross-Encoder feeds query and passage together through full self-attention, dramatically reducing false-positive passage retrieval.
- **Measured Metric:** Hybrid sparse-dense retrieval with RRF significantly outperforms dense-only retrieval on technical queries with domain acronyms. (89.4% vs 71.2% (FAISS Dense Alone)).

NextSense: Sequential Recommendation Engine

PyTorch, Python, Polars, CUDA

Sept. 2026 – Nov. 2026

Deep sequential recommendation system modeling temporal user interaction trajectories using recurrent and self-attentive backbones for high-recall candidate generation.

- **Causal Self-Attention over Recurrent Architectures (GRU):** Recurrent networks compress the entire sequence history into a single fixed-size hidden vector h_t , causing gradient decay on sessions with 30+ interactions. Causal self-attention dynamically attends to any past interaction regardless of temporal distance.
- **In-Batch Negative Sampling with Frequency Weighting:** Computing exact softmax over 50,000+ items per sequence step is prohibitively expensive during training. Sampling negative items uniformly from other users within the same minibatch provides strong negative signals with zero additional memory lookup.
- **Measured Metric:** Sequential Transformer modeling (SASRec) significantly outperforms static collaborative filtering on session-based item recommendation. (34.2% vs 14.8% (BPR Matrix Factorization)).

EXPERIENCE & ACADEMIC APPOINTMENTS

Teaching Assistant — Introduction to Robotics

Jul 2026 – Nov 2026

Indian Institute of Technology (IIT) Guwahati

- Assisted in delivering laboratory sessions covering forward and inverse kinematics, path planning algorithms, and rigid-body transforms.
- Mentored undergraduate students on ROS / Python implementations and graded coursework and laboratory demonstrations.

Technology Analyst — Cybersecurity & Identity Governance

2024 – 2025

Fiserv • Global Technology Solutions

- Engineered and automated identity governance and role-based access control (RBAC) schemas in SailPoint IdentityIQ for 40,000+ enterprise accounts in regulated fintech environments.
- Reduced compliance auditing overhead by 35% through automated certification workflows and identity lifecycle pipelines.
- Hardened internal banking API endpoints against automated Denial-of-Service (DoS) and account takeover vectors via rate limiting and session validation logic.

Cybersecurity Engineering Intern

2023

Fiserv

- Automated identity connector reconciliations using Java and REST APIs, reducing synchronization failure rates.
- Conducted vulnerability assessments and built telemetry dashboards for security access logs.

TECHNICAL SKILLS & DOMAIN EXPERTISE

ML Architectures & Math: Transformers, Diffusion Models, RoPE, RMSNorm, SwiGLU, KV-Cache, DPO, DDPM, DDIM, CFG, SASRec, RRF, Cross-Encoders	Languages & Frameworks: Python, PyTorch, C++, CUDA (basic), TypeScript, Next.js, React, Java
Infrastructure & Systems: FAISS, BM25, Linux Kernel, Git, Docker, HuggingFace Transformers & Diffusers, ONNX	Systems Evaluation: Ablation Studies, Latency Profiling (torch.profiler), VRAM Allocation Ceilings, NLI

